

From Homogeneous to Heterogeneous

A methodology for operator learning
in realistic contaminant transport

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Where we are

— Four reports, one unified study, near-perfect results —

Four reports in five months

Dec 2025

PINN Baseline

Single-Pe homogeneous
Proof-of-concept

Jan 2026

Parametric PINN

$\log Pe$ in $[1, 10^5]$
One model, all regimes

Mar 2026

Neural Operator

DeepONet branch+trunk
Ogata-Banks reference

Apr 2026

CFL-Unified

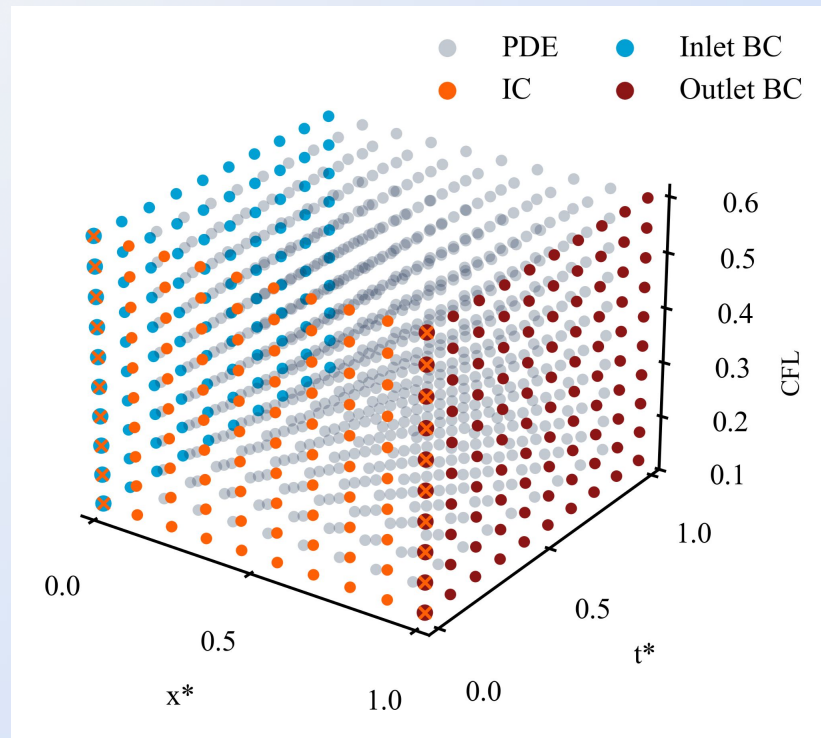
Comsol-verified, LBFGS,
regular mesh

By April we have three working surrogates — PINN, Parametric PINN, PINO — all trained under one protocol.

The CFL-unified study (April 2026 report)

Physical Setup

L	100 m	domain length
T_{max}	1200 d	time horizon
D	$10^{-6} \text{ m}^2/\text{s}$	fixed diffusion
U	0.01 - 0.05 m/d	varying velocity
CFL	0.12 - 0.60	$UT_{\max} / L$
Pe	0.010368	$DT_{\max} / L^2$

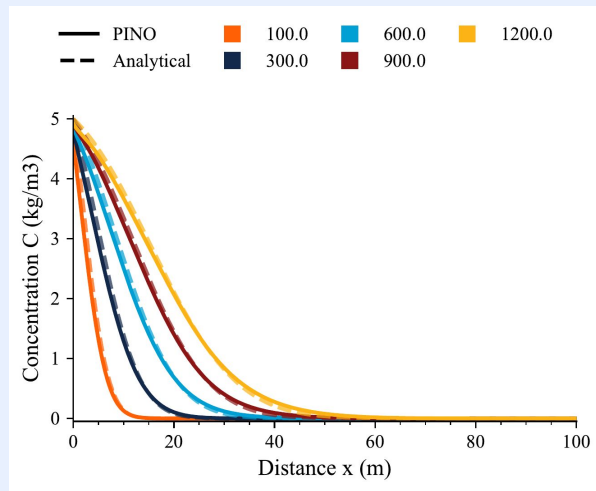


Regular 50^3 collocation mesh in (x^*, t^*, CFL)

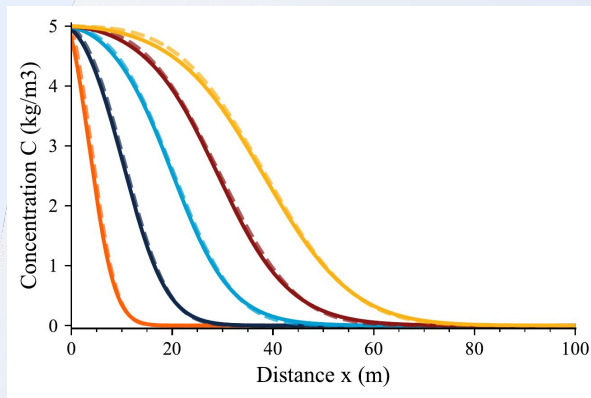
Unified across all three models: **L-BFGS** + **regular mesh**. Only architecture & how CFL enters differ.

Near-perfect across the board — PINO at three CFLs

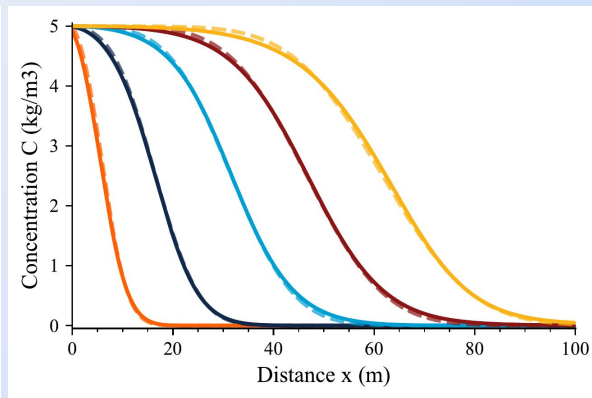
Single trained operator, swept from diffusion-shallow to advection-sharp. Ogata-Banks = analytical reference.



CFL = 0.12
diffusion-domination



CFL = 0.36
mixed regime



CFL = 0.60
advection-domination

✓ Ogata-Banks validated against COMSOL for the same five CFLs — analytical reference holds.

The homogeneous problem is **done**.

Three surrogates, one protocol, near-perfect agreement with analytics and COMSOL.

3 models

PINN · Parametric PINN · PINO

5 CFLs

0.12 → 0.60 in one shot

Near-perfect

vs Ogata-Banks & COMSOL

Ready for the real challenge: **heterogeneous** media

The heterogeneous challenge

- Real aquifers are not homogeneous, the parameterization must grow —

In real aquifers **U** varies in space

Permeability, porosity, and fracture patterns make $U(x)$ a function, not a scalar.

Up to now — **homogeneous**

$$\partial C / \partial t + U \partial C / \partial x = D \partial^2 C / \partial x^2$$

U, D are scalars $\rightarrow$ one CFL captures the whole family.

- ✓ Scalar CFL fits as branch input
- ✓ Cheap data: Ogata–Banks analytic
- ✓ Solved — near-perfect results

The next step — **heterogeneous**

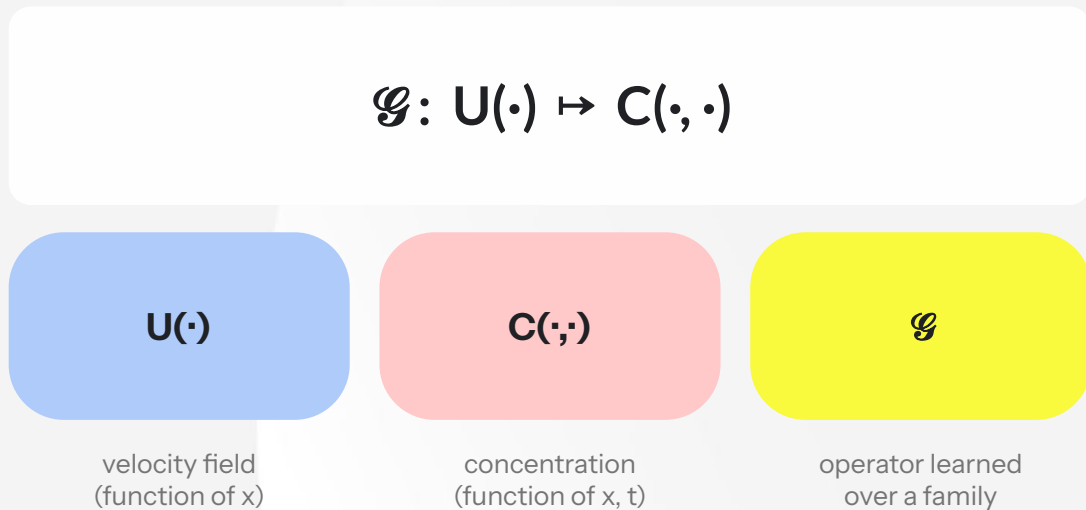
$$\partial C / \partial t + \mathbf{U(x)} \partial C / \partial x = D \partial^2 C / \partial x^2$$

$U(x)$ is a function $\rightarrow$ no scalar CFL can cover it.

- ✗ Parametric PINN input explodes
- ✗ No analytic solution
- $\rightarrow$ Need an **operator surrogate**

The operator view — already on the table

We introduced this framing back in February. Now we make it operational.



Goal: train **one surrogate** that, given a new velocity field, predicts the full concentration evolution — **no retraining**.

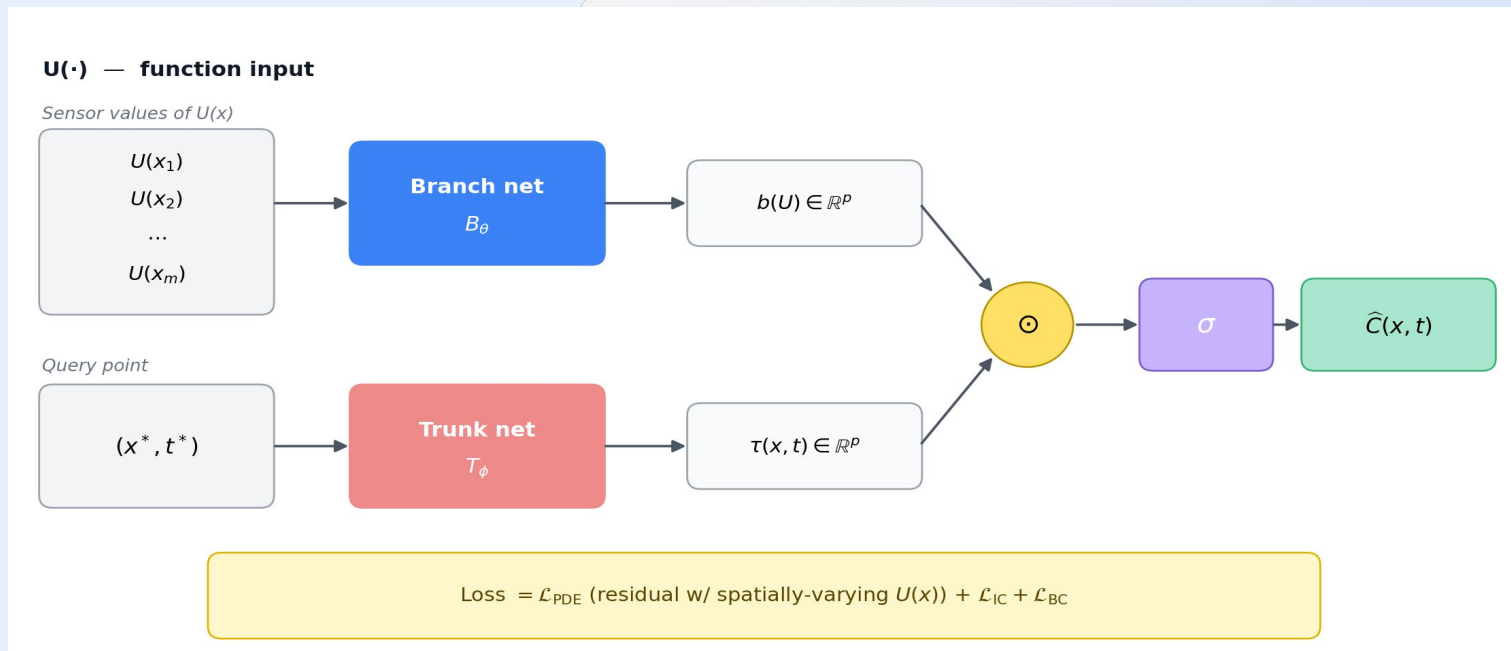


Proposed methodology

- Extend the DeepONet branch from a scalar to a function —

Core idea — functional branch input

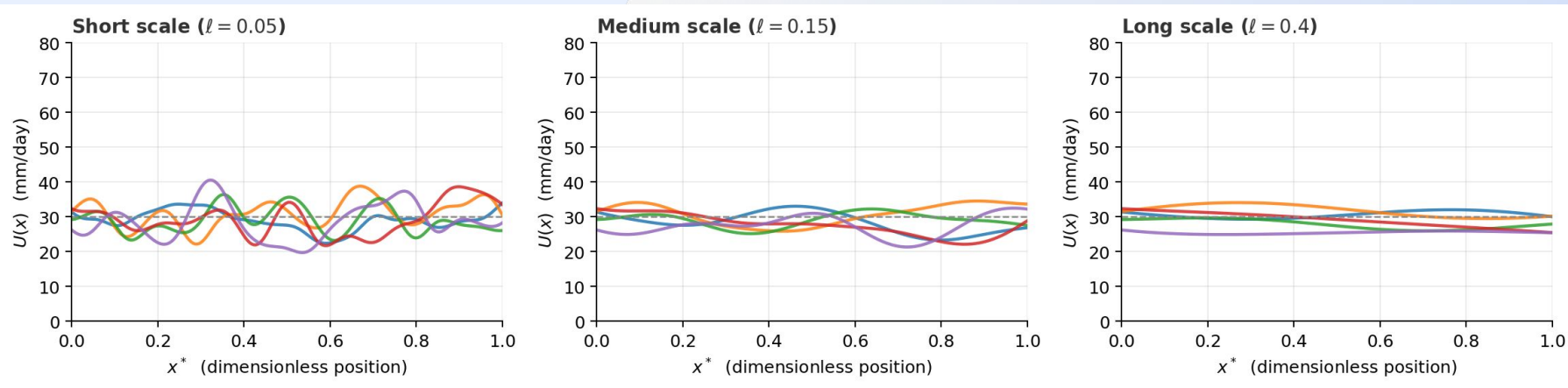
Keep the DeepONet architecture we already have. Replace scalar CFL with $U(x)$ sampled at m sensors.



Minimal change: swap one scalar for a **sensor vector** of length m , reuse the same branch+trunk+dot-product machinery.

Training inputs – Gaussian random fields

Sample diverse $U(x)$ from a GRF with controlled correlation length; train physics-only on each sample.



Sample $U(x)$

GRF with correlation length ℓ and variance σ^2 (KL expansion)

Train (physics-only)

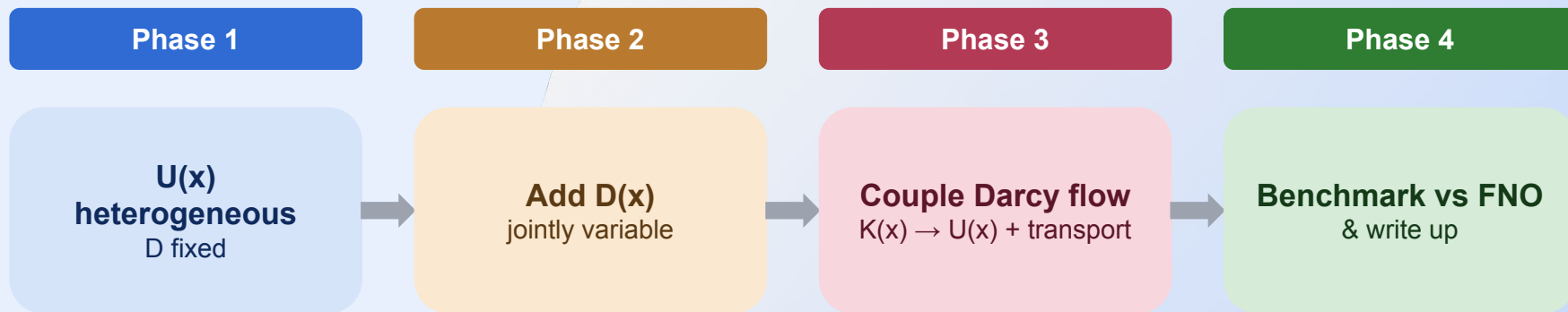
PDE residual + IC + BC, evaluated at collocation points; no labels

Verify with COMSOL

Post-training: solve the ADE on held-out $U(x)$ and compare against the model

Staged rollout

Each phase reuses the infrastructure of the previous one.



Alternatives briefly considered

FNO (larger data, spectral bias) · UNet (mesh-dependent) · GNN (over-engineered for 1D) → PINO/DeepONet is the minimum viable extension.

Open questions — for today

Decisions that shape Phase 1 — your input would be valuable.

1

GRF statistics

Correlation length ℓ range, variance σ^2 ? Physically meaningful ranges for aquifer $U(x)$?

2

Sensor count m

How many branch sensors to fix? Rule of thumb: $m \gtrsim 4 \cdot (L/\ell)$. Start with $m = 40$?

3

Verification solver

Stick with COMSOL for verification, or build an in-house finite-volume reference for faster iteration?

4

Generalization scope

Train and test on the same GRF, or include polynomial / Fourier OOD test families?

Thank you

